

CURRICULUM VITAE ABREVIADO (CVA)

IMPORTANT – The Curriculum Vitae cannot exceed 4 pages. Instructions to fill this document are available in the website.

Part A. PERSONAL INFORMATION *

First name	Antonio José		
Family name	Cuesta Vázquez		
Gender (*)	Male	Birth date (dd/mm/yyyy)	
Social Security, Passport, ID number			
e-mail	ajcuesta@uco.es	URL Web	https://www.uco.es/~ajcuesta
Open Researcher and Contributor ID (ORCID) (*)	0000-0002-4153-9470		

(*) Mandatory

A.1. Current position

Position	Profesor Titular de Universidad		
Initial date	16/10/2023		
Institution	Universidad de Córdoba		
Department/Center	Departamento de Física	Facultad de Ciencias	
Country	Spain	Teleph. number	957212064
Key words	Cosmology, Large Scale Structure of the Universe, Galaxy Redshift Surveys, Baryon Acoustic Oscillations, Dark Energy, Dark Matter, Neutrino Cosmology, Bayesian Inference, Monte Carlo Methods		

A.2. Previous positions (research activity interruptions, indicate total months)

Period	Position/Institution/Country/Interruption cause
2022-2023	Profesor Contratado Doctor. Universidad de Córdoba, Spain
2018-2022	Profesor Ayudante Doctor. Universidad de Córdoba, Spain
2017-2018	Profesor Sustituto Interino. Universidad de Córdoba, Spain
2013-2017	Postdoctoral Researcher. Institut de Ciències del Cosmos, Universitat de Barcelona (ICCUB), Spain
2010-2013	Postdoctoral Research Associate. Yale Center for Astronomy and Astrophysics, Yale University, United States of America
2006-2010	FPU grant. Instituto de Astrofísica de Andalucía (IAA-CSIC), Spain

(Include all the necessary rows)

A.3. Education

PhD, Licensed, Graduate	University/Country	Year
Licenciado en Física	Universidad de Granada, Spain	2005
Doctor en Física	Universidad de Granada, Spain	2010

(Include all the necessary rows)

Part B. CV SUMMARY (max. 5000 characters, including spaces)

After completing my BSc in Physics at Universidad de Granada (2005), I started my graduate studies at Instituto de Astrofísica de Andalucía (IAA-CSIC) under the supervision of Francisco Prada. There I obtained a FPU grant that allowed me to work on the analysis of N-body cosmological simulations. My PhD thesis "The properties of virialized dark matter halos: signals from the large scale structures" (2010) focused on the definition of dark matter halo radius and other properties such as the indirect detection of dark matter halos.

The definition of dark matter halo radius had the biggest impact on the community. During my PhD research visit to Harvard-Smithsonian Center for Astrophysics (2009), I could work with researchers involved in that same topic. But it was my PhD research visit to Lawrence Berkeley

National Laboratory (2008) which opened my doors to become involved with Sloan Digital Sky Survey (SDSS) researchers, which earned myself a postdoctoral position at Yale Center for Astronomy and Astrophysics (2010-2013). Under the mentorship of Nikhil Padmanabhan, I could work with him and with Daniel Eisenstein (who lead the discovery of baryon acoustic oscillations (BAO) in 2005 using SDSS data) on the analysis of the Baryon Oscillation Spectroscopic Survey (BOSS).

During that first postdoctoral position I became heavily involved in the Galaxy Clustering working group. Although I worked on very different analysis of the galaxy clustering data, it was my expertise on running reconstruction algorithms on galaxy catalogs to increase the signal-to-noise in BAO measurements, as well as running cosmological MCMC codes such as CosmoMC (I even wrote a tutorial that became quite popular at that moment), as well as my contribution to the first 2D BAO measurement which earned another postdoctoral position, this time at Institut de Ciències del Cosmos at Universitat de Barcelona (2013-2016), under the mentorship of Licia Verde.

In this second postdoctoral position, I could continue contributing to the Galaxy Clustering working group of the BOSS collaboration, as well as take on new projects in parallel with it. While I lead my analysis of BAO in DR12 LOWZ and CMASS galaxies, I could work on the cosmological analysis of Horndeski theories, the cosmological determination of neutrino mass, and even on the analysis of the cosmological distance ladder (later dubbed "The Hubble Tension"), in a paper with Nobel Laureate Adam Riess.

After these two fruitful postdoctoral positions I co-authored 53 SDSS research papers. As of now, I have co-authored a total of 74 papers (accounting for more than 18,000 citations), out of which 8 are 1st author papers, 6 are 2nd author papers, and 12 are 3rd author papers, with a h-index of 52 (Web Of Science).

Back in Andalusia, I then obtained a tenure-track position at Universidad de Cordoba (2018), my current institution. Here I started teaching Physics degree courses, such as Atomic & Molecular Physics, and Nuclear & Particle Physics. Eventually I was promoted to a permanent position at Universidad de Cordoba (Profesor Titular since 2023).

My most recent research topic since 2018, with colleagues at Universidad de Granada, is the study of constraints on the axion-like majoron model from cosmological datasets such as CMB, BBN, and LSS, in the context of cosmological tensions. More recently, I also became the coordinator of the Theoretical Cosmology group at the J-PAS collaboration using Spanish ICTS Observatorio Astrofísico de Javalambre. I also co-authored two papers on Nature Astronomy with colleagues from Beijing universities; both were featured at Universidad de Cordoba's YouTube channel.

Since 2020 I was held responsible of teaching Astrophysics & Cosmology. Thanks to a sabbatical (Ayudas para la Recualificación del Sistema Universitario Español) in 2023 at Universidad de Granada, I started co-authoring a book on solved problems in Astrophysics that was published in early 2026. As part of my teaching duties at Universidad de Cordoba, I mentored 18 Bachelor Thesis on Astrophysics and Cosmology. I also participated in 3 teaching innovation projects such as UCOsmos, in which I co-hosted an Astrophysics show on Twitch with students from Universidad de Cordoba. I regularly participate in outreach events such as Paseo por la Ciencia or Noche Europea de I@s Investigador@s (Researchers' Night).

Since I started working at Universidad de Cordoba (UCO), I have actively worked towards the consolidation of Astrophysics at UCO towards the attraction of highly talented colleagues to tenure-track and postdoctoral positions, and the hiring of two postdoctoral researchers with funds from Planes Complementarios en Astrofísica y Física de Altas Energías (116 083 euros, PI at UCO: Antonio J. Cuesta). It is a fact that Astrophysics at Universidad de Cordoba is now stronger than ever before, with more than 25 publications in the last 5 years, and 6 people working actively on Astrophysics and Cosmology.

Part C. RELEVANT MERITS (sorted by typology)

C.1. Publications (see instructions)

Qin, Fuxing; Wang, Yuting (CA); Zhao, Gong-Bo; ... ; Vázquez Ramió, Héctor, including Antonio J. Cuesta (4/31) (2026). J-PAS and PFS Surveys in the Era of Dark Energy and Neutrino Mass Measurements. *The Astrophysical Journal*. 997, 251, pp. 1-14.

Rodrigues, Gabriel (CA); Antonio J. Cuesta (2/33); Alcaniz, Jailson; ... ; Vázquez Ramió, Héctor (2025). J-PAS: forecasting constraints on neutrino masses. *Journal of Cosmology and Astroparticle Physics*. 12, 047, pp. 1-25.

Sanz-Wuhl, Santiago; Gil-Marín, Héctor (CA); Antonio J. Cuesta; Verde, Licia (2024). BAO cosmology in non-spatially flat background geometry from BOSS+eBOSS and lessons for future surveys. *Journal of Cosmology and Astroparticle Physics*. 05, pp. 1-41.

Xu, Kun; Jing, Y. P. (CA); Zhao, Gong-Bo; Antonio J. Cuesta (2023). Evidence for baryon acoustic oscillations from galaxy-ellipticity correlations. *Nature Astronomy*. 7, pp. 1259-1264.

Antonio J. Cuesta; Illana-Calero, José Ignacio (CA); Masip, Manuel (2023). Photon to axion conversion during Big Bang Nucleosynthesis. *Journal of Cosmology and Astroparticle Physics*. 11, pp. 1- 11.

Antonio J. Cuesta; Gómez-Santamaría, Mario Emilio; Illana-Calero, José Ignacio; Masip, Manuel (CA) (2022). Cosmology of an axion-like majoron. *Journal of Cosmology and Astroparticle Physics*. 04, pp. 1-17.

Zhao, Gong-Bo (CA); Raveri, Marco; Pogosian, Levon; ... ; Zhang, Hanyu, including Antonio J. Cuesta (11/26) (2017). Dynamical dark energy in light of the latest observations. *Nature Astronomy*. 1, pp. 627-632.

Alam, Shadab; Ata, Metin; Bailey, Stephen; ... ; Gong-Bo Zhao (CA: Beth A. Reid and Jeremy L. Tinker), including Antonio J. Cuesta (12/73) (2017). The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample. *Monthly Notices of the Royal Astronomical Society*. 470, pp. 2617-2652.

Antonio J. Cuesta (CA); Niro, Viviana; Verde, Licia (2016). Neutrino mass limits: robust information from the power spectrum of galaxy surveys. *Physics of the Dark Universe*. 13, pp. 77-86.

Antonio J. Cuesta (CA, 1/23); Vargas-Magaña, Mariana; Beutler, Florian; ... ; Martin White (2016). The clustering of galaxies in the SDSS-III Baryon Oscillation Spectroscopic Survey: baryon acoustic oscillations in the correlation function of LOWZ and CMASS galaxies in Data Release 12. *Monthly Notices of the Royal Astronomical Society*. 457, pp. 1770-1785.

Antonio J. Cuesta (CA); Verde, Licia; Riess, Adam; Jiménez, Raúl (2015). Calibrating the cosmic distance scale ladder: the role of the sound horizon scale and the local expansion rate as distance anchors. *Monthly Notices of the Royal Astronomical Society*. 448, pp. 3463-3471.

C.2. Congress, indicating the modality of their participation (invited conference, oral presentation, poster)

Oral presentation. Non-cold relics in CLASS. Antonio J. Cuesta. Iberian Cosmology Meetings (IberiCos 2026). Granada 22/04/2026-24/04/2026.

Oral presentation. LIA4@UCO: from neutrino masses to dark energy. Antonio J. Cuesta. Reunión Final del Plan Complementario de Astrofísica y Altas Energías. Santander 25/11/2025-27/11/2025.

Oral presentation. Cosmological implications of an Axion-Like Majoron. Antonio J. Cuesta. XI Meeting on fundamental Cosmology. Santander 18/11/2025-20/11/2025.

Invited conference. Cosmological implications of an Axion-Like Majoron. Antonio J. Cuesta. Tensions in Cosmology. Corfú (Grecia) 2/09/2025-8/09/2025.

Oral presentation. Forecasts on neutrino masses with J-PAS. Antonio J. Cuesta. XX J-PAS Collaboration Meeting. Teruel 18/11/2024-20/11/2024.

Oral presentation. The H0 tension and the physics of the neutrino sector. Antonio J. Cuesta. Tensions in Cosmology. Corfú (Grecia) 7/09/2022-12/09/2022.

C.3. Research projects, indicating your personal contribution. In the case of young researchers, indicate lines of research for which they have been responsible.

AST22_00001. Análisis de la estadística de la distribución de galaxias en la estructura a gran escala mediante grandes cartografiados astronómicos. Junta de Andalucía. Consejería de Universidad, Investigación e Innovación. 2021-2025. 116,083 EUR. Principal Investigator (PI: Antonio J. Cuesta, Universidad de Córdoba).

PID2022-140440NB-C21. Física de Astropartículas y QCD. Agencia Estatal de Investigación. Ministerio de Ciencia e Innovación. 2023-2026. 153,250 EUR. Researcher (PI: Manuel Masip, Universidad de Granada).

PID2019-107844GB-C21. Física de Astropartículas y QCD. Agencia Estatal de Investigación. Ministerio de Ciencia e Innovación. 2020-2023. 89,540 EUR. Researcher (PI: Manuel Masip, Universidad de Granada).

AYA2014-58747-P. Cosmología Física en la época de grandes cartografiados. Ministerio De Economía y Competitividad. 2015-2018. 227,480 EUR. Researcher (PI: Licia Verde, Institut de Ciències del Cosmos, Universitat de Barcelona).

NNX11AF43G. Reconstructing baryon oscillations: Enhancing the dark energy reach of future redshift surveys. NASA. 2011-2015. Researcher (PI: Nikhil Padmanabhan, Yale Center for Astronomy and Astrophysics, Yale University).

FP7-IDEAS-Phys.LSS-240117. Cosmological Physics with future large-scale structure surveys. European Research Council. 2009-2015. 1,395,000 EUR. Researcher (PI: Licia Verde, Institut de Ciències del Cosmos, Universitat de Barcelona).

C.4. Contracts, technological or transfer merits, Include patents and other industrial or intellectual property activities (contracts, licenses, agreements, etc.) in which you have collaborated. Indicate: a) the order of signature of authors; b) reference; c) title; d) priority countries; e) date; f) Entity and companies that exploit the patent or similar information, if any.